

Traumatic Panniculitis

Abelardo Moreno, MD^a, Joaquim Marcoval, MD^b,
Jordi Peyri, MD^{b,*}

KEYWORDS

• Traumatic panniculitis • Factitial panniculitis

Traumatic panniculitis includes changes in the subcutaneous fat secondary to both physical and chemical agents. The causes of physical injuries are trauma, cold, and electricity. Cold panniculitis is described elsewhere in this issue. The so-called “equestrian cold panniculitis” is caused by a combination of cold exposure and equestrian activity.¹ In general, in traumatic panniculitis, the intensity of traumatic injury is not necessary related to the development or the intensity of the cutaneous lesions. History of trauma helps in the diagnosis, although often the antecedent is minor or even unnoticed. Traumatic fat necrosis can occur in patients of all ages. In adults it is most commonly seen as breast masses in women. Fat necrosis can also be of factitial origin. Acute presentation as a large mass is quite unusual.² Repeated micro-traumas is considered the cause of semicircular lipoatrophy³ that can also be considered a particular form of traumatic panniculitis. Posttraumatic subcutaneous nodules can also be totally or nearly totally encapsulated by thin, fibrous tissue. Such names as “nodular-cystic fat necrosis,” “mobile encapsulated lipoma,” and “encapsulated necrosis” have been offered to designate this particular form.⁴

In electrical injury, the lesions are located at the point of contact. Sometimes the use of electrodes, medical electrostimulation equipment, or acupuncture may cause skin burns with superficial subcutaneous involvement.

The term “factitial panniculitis” refers to subcutaneous reactions secondary to injection of substances. All types of substances can be injected in the skin producing similar reactions.^{5,6} A peculiar variant of factitial panniculitis is sclerosing lipogranuloma (paraffinoma) secondary to

injection of exogenous oily substances sometimes for cosmetic purposes. In cutaneous biopsies, oil deposits, necrosis of the subcutaneous fat, and a secondary local sclerosing reaction characterize paraffinoma.

CLINICAL FEATURES

The clinical picture of traumatic panniculitis is non-specific, although the clinical distribution of the lesions may suggest an exogenous origin. The cutaneous lesions appear as indurated, warm, red, subcutaneous plaques or nodules. In cases of extensive liquefaction it may be discharged through superficial wound (**Fig. 1**).⁷ In electric injury panniculitis is also a superficial burn at the point of injury. In factitial panniculitis the cutaneous lesion appears at the site of injury as painful indurated nonspecific plaques and nodules. Superficial marks of injection may help in diagnosis. Patients with repeated injection can follow a clinical course of exacerbations and remission that probably in the past could be diagnosed as Weber-Christian disease.

HISTOPATHOLOGY

The histologic picture of traumatic panniculitis is not specific, although late lesions are quite characteristic. Early lesions are characterized by an inflammatory infiltrate composed of lymphocytes and macrophages starting around vessels and septa (**Fig. 2**). As the lesion advances, there is associated rupture of fat cells and formations of fat microcysts surrounded by histiocytes (**Fig. 3**). There are also collections, usually small, of foam cells together with inflammatory cells that can

^a Department of Pathology, Hospital Universitari de Bellvitge, IDIBELL, Barcelona, Spain

^b Department of Dermatology, Hospital Universitari de Bellvitge, IDIBELL, Barcelona, Spain

* Corresponding author. Dermatopathology, Department of Pathology, Hospital Universitari de Bellvitge, Feixa Llarga s/n, 08907 L'Hospitalet LI, Barcelona, Spain.

E-mail address: jpeyri@csb.scs.es (J. Peyri).



Fig. 1. Nonspecific indurated, red, subcutaneous plaques and ulcerated lesions in the leg secondary to trauma.

include neutrophils and eosinophils. Late lesions show fibrotic replacement of damaged fat and residual small fat cysts surrounded by macrophages and foreign body-type giant cells (**Fig. 4**). Lipomembranous changes (small cystic spaces lined by amorphous eosinophilic material with small intraluminal projections) have also been described (7 of 34 cases in the series of traumatic panniculitis from Diaz-Cascajo and Borghi).⁸ Finally, dystrophic calcic deposits may be seen in late, residual lesions.

In nodular-cystic fat necrosis, the histopathologic examination discloses necrosis of fat tissue similar to that described and even including pseudomembranous change⁹ surrounded by a fibrous capsule that delimited the lesions and gives a pseudotumoral appearance.

The histologic picture of early lesions of factitial panniculitis is characterized by a lobular or mixed suppurative panniculitis nonspecific for diagnosis. Late lesions show a picture similar to traumatic panniculitis with necrotic fat cells, macrophages,

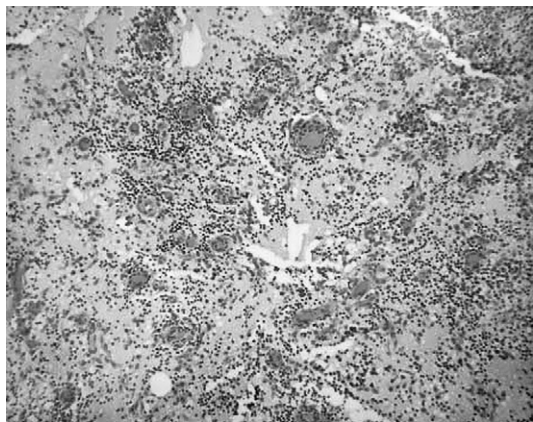


Fig. 2. Early lesion of traumatic panniculitis showing a nonspecific inflammatory infiltrate of lymphocytes and macrophages. Hematoxylin-eosin stain, original magnification $\times 150$.

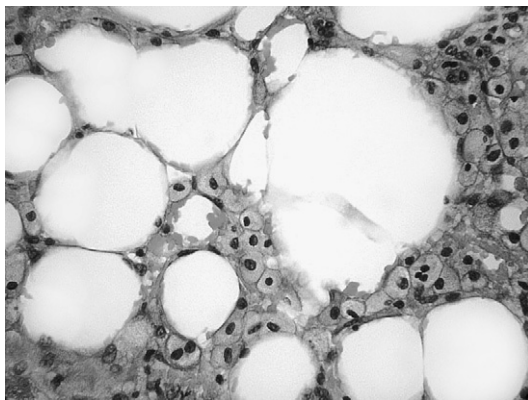


Fig. 3. Fat pseudocysts related to fat cell rupture, surrounded by histiocytes. Hematoxylin-eosin stain, original magnification $\times 300$.

and inflammatory infiltrate. The hallmark for diagnosis is the presence of macrophages that contain refractile foreign material, although depending on the injected substance there is no foreign body reaction. In paraffinoma large fat deposits gives the tissue a Swiss cheese appearance with secondary fibrosis.

TREATMENT

Traumatic panniculitis is usually a self-limiting disorder and requires only symptomatic relief. Some patients require surgical treatment of local complications (indurated fibrous masses or large cysts or nodules). Early and serial debridement of necrotic tissue is the treatment of preference for electric burns.¹⁰ The treatment of factitial panniculitis may require psychiatric support.

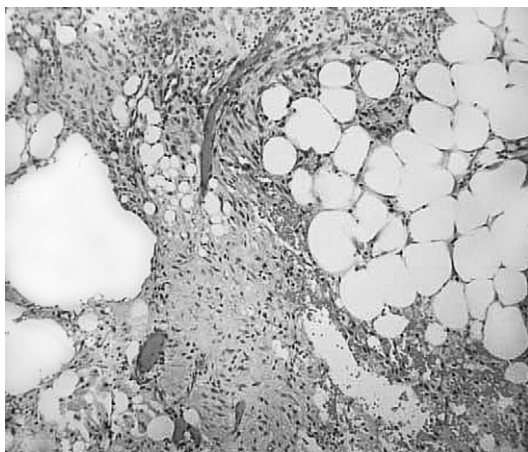


Fig. 4. Late lesion of traumatic panniculitis with fibrotic replacement of damaged fat and residual fat cysts surrounded by macrophages and foreign body-type reaction. Hematoxylin-eosin stain, original magnification $\times 150$.

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